

Financial Econometrics - Homework

Chapter 15

Panel Data Models

Exercise 15.6

15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10 Estimation Results for Exercise 15.6

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
<i>C</i>	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
<i>EXPER</i>	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
<i>EXPER</i> ²	−0.0033 (0.0011)	−0.0031 (0.0011)	−0.0012 (0.0011)	−0.0012 (0.0011)	−0.0023 (0.0007)
<i>SOUTH</i>	−0.2128 (0.0338)	−0.2384 (0.0344)	−0.3261 (0.1258)	−0.3261 (0.2495)	−0.2326 (0.0317)
<i>UNION</i>	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
<i>N</i>	716	716	1432	1432	1432

(standard errors in parentheses)

- The OLS estimates of the $\ln(WAGE)$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?
- The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it}) \quad (XR15.6)$$

- Explain any differences in assumptions between this model and the models in part (a).
- Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?
- The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.
- Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?
- Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

a. Comparison of OLS estimates for 1987 and 1988.

The OLS results for 1987 and 1988 are very similar (e.g., EXPER is 0.1270 vs 0.1265, SOUTH is -0.2128 vs -0.2384). When we estimate these cross-sections separately, we assume that the parameters do not vary across individuals and that there is no unobservable individual heterogeneity correlated with the explanatory variables (i.e., any unobserved factors are randomly distributed and absorbed in the error term).

b. Differences in assumptions in the Panel Data Model.

The panel data regression model (XR15.6) explicitly includes an unobserved, time-invariant individual heterogeneity term, u_i . Unlike the OLS models in (a), the panel model explicitly accounts for individual differences (like innate ability). Depending on the approach (Fixed Effects vs Random Effects), the panel model either allows u_i to be correlated with the explanatory variables (FE) or assumes it is strictly uncorrelated and exploits it for efficiency (RE).

c. Comparison between Fixed Effects (FE) and OLS.

Comparing Column (3) FE to Columns (1) and (2) OLS: The most significant differences are observed in the coefficients for EXPER (dropping from ~ 0.127 to 0.0575) and SOUTH (dropping from ~ -0.21 to -0.3261). This suggests that OLS suffered from omitted variable bias, as unobserved characteristics (like ability) were likely correlated with these variables.

d. F-test for no individual differences.

```
15 # d. F-test degrees of freedom and critical value
16 # N = 716, T = 2, NT = 1432. Ks = 4 (EXPER, EXPER2, SOUTH, UNION)
17 N <- 716
18 T_obs <- 2
19 Ks <- 4
20 df1 <- N - 1
21 df2 <- N*T_obs - N - Ks
22
23 F_stat_156 <- 11.68
24 F_crit_156 <- qf(0.99, df1, df2)
25
26 cat("15.6d: F-test for no individual differences\n")
27 cat("df1: ", df1, "\ndf2: ", df2, "\n")
28 cat("1% Critical Value: ", F_crit_156, "\n\n")
```

The F-statistic is 11.68. For a fixed effects model, the degrees of freedom are $df1 = N - 1 = 716 - 1 = 715$, and $df2 = NT - N - Ks = 1432 - 716 - 4 = 712$. The 1% critical value for $F(715, 712)$ is 1.19. Since $11.68 > 1.19$, we strongly reject the null hypothesis of no individual differences. Individual heterogeneity is present.

e. Fixed Effects with Cluster-Robust Standard Errors.

Column (4) uses cluster-robust standard errors, which relax the assumptions of homoskedasticity and no serial correlation within an individual's observations across time. Comparing to Column

(3), the standard error for SOUTH increases substantially (0.1258 to 0.2495). The robust standard errors are generally larger, reflecting within-group correlation or heteroskedasticity.

f. Random Effects vs Fixed Effects and Hausman Test.

```
30 # f. Hausman t-test for specific coefficients
31 # t = (b_FE - b_RE) / sqrt(se_FE^2 - se_RE^2)
32 calc_hausman_t <- function(b_fe, b_re, se_fe, se_re) {
33   var_diff <- se_fe^2 - se_re^2
34   if (var_diff <= 0) return(NA)
35   return((b_fe - b_re) / sqrt(var_diff))
36 }
37
38 cat("15.6f: Hausman t-tests\n")
39 t_exper <- calc_hausman_t(0.0575, 0.0986, 0.0330, 0.0220)
40 t_south <- calc_hausman_t(-0.3261, -0.2326, 0.1258, 0.0317)
41 t_union <- calc_hausman_t(0.0822, 0.1027, 0.0312, 0.0245)
42
43 cat("t-stat EXPER: ", t_exper, "\n")
44 cat("t-stat SOUTH: ", t_south, "\n")
45 cat("t-stat UNION: ", t_union, "\n\n")
46
```

Comparing RE (Col 5) to FE (Col 3), the coefficients for EXPER (0.0986 vs 0.0575) and SOUTH (-0.2326 vs -0.3261) show the most difference. We compare RE with the non-robust FE (Col 3) because the Hausman test formulation assumes the RE estimator is fully efficient under the null, which relies on the classical homoskedastic standard errors.

Hausman t-tests for specific coefficients (calculated in R):

- t-stat EXPER: -1.671

- t-stat SOUTH: -0.768

- t-stat UNION: -1.061

None of these individual t-statistics exceed the 5% critical value (1.96). While individual tests don't strictly reject exogeneity, the differences in EXPER are sizable. Often, a joint Hausman test is preferred, but based on these individual tests, RE cannot be definitively ruled out, though FE is generally safer when correlating unobserved ability with experience is suspected.

Exercise 15.20

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?
- Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]
- Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?
- Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.
- Using the *t*-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?
- Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

a. Pooled OLS Regression.

```
50 cat("--- Exercise 15.20 ---\n")
51 data("star")
52
53 # Create a pdata.frame. The "group" is schid, and the "time" is id.
54 # This represents clustering of students within schools.
55 star_p <- pdata.frame(star, index = c("schid", "id"))
56
57 # a. Pooled OLS
58 mod_pool <- plm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
59               data = star_p, model = "pooling")
60 cat("15.20a: Pooled OLS Model\n")
61 print(summary(mod_pool))
62
```

The pooled OLS estimation yields the following conclusions: Students in small classes perform better (coef: 5.82, $p < 0.001$). Teacher aides do NOT significantly improve scores ($p = 0.39$). Students of more experienced teachers score slightly higher (coef: 0.49, $p < 0.001$). Boys score lower on reading (-6.15, $p < 0.001$), White/Asian students score higher (+3.90, $p < 0.001$), and students on free lunch score lower (-14.77, $p < 0.001$).

b. School Fixed Effects.

```

62
63 # b. School Fixed Effects
64 mod_fe <- plm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
65             data = star_p, model = "within")
66 cat("\n15.20b: School Fixed Effects Model\n")
67 print(summary(mod_fe))
68

```

Using school fixed effects, the coefficients change slightly but the overall conclusions remain the same. The effect of small classes increases slightly to 6.49 ($p < 0.001$), teacher experience drops to 0.28 ($p < 0.001$), and the White/Asian premium increases to 8.02. This means that even controlling for unobserved school-level characteristics, small classes are beneficial.

c. Test for significance of school fixed effects.

```

68
69 # c. Test for significance of school fixed effects
70 fe_test <- pFtest(mod_fe, mod_pool)
71 cat("\n15.20c: F-test for School Fixed Effects\n")
72 print(fe_test)
73

```

The F-test comparing the FE model to the Pooled model yields $F = 16.698$ ($p < 0.001$), meaning school fixed effects are highly significant. We would expect the inclusion of fixed effects to have little influence on the remaining coefficients ONLY IF the unobserved school effects are uncorrelated with the explanatory variables (e.g., if students and teachers were randomly assigned ACROSS schools, not just within them).

d. School Random Effects and LM Test.

```

74 # d. School Random Effects
75 mod_re <- plm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
76             data = star_p, model = "random")
77 cat("\n15.20d: School Random Effects Model\n")
78 print(summary(mod_re))
79
80 # LM test for random effects
81 lm_test <- plmtest(mod_pool, effect = "individual", type = "bp")
82 cat("\n15.20d: LM Test for Random Effects\n")
83 print(lm_test)

```

The Random Effects (RE) model yields coefficients very similar to the FE model. The Lagrange Multiplier (LM) test for random effects gives a Chi-squared statistic of 6677.4 ($p < 0.001$), strongly rejecting the null hypothesis of no random effects. This confirms there is significant unobserved heterogeneity at the school level.

e. Hausman t-tests for Endogeneity.

```

85 # e. Hausman t-tests for coefficients
86 cat("\n15.20e: Hausman t-tests for specific coefficients\n")
87 coef_fe <- coef(mod_fe)
88 coef_re <- coef(mod_re)
89 se_fe <- sqrt(diag(vcov(mod_fe)))
90 se_re <- sqrt(diag(vcov(mod_re)))
91
92 vars_to_test <- c("small", "aide", "tchexper", "white_asian", "freelunch", "boy")
93
94 for (var in vars_to_test) {
95   if (var %in% names(coef_fe) && var %in% names(coef_re)) {
96     b_fe <- coef_fe[var]
97     b_re <- coef_re[var]
98     s_fe <- se_fe[var]
99     s_re <- se_re[var]
100
101     var_diff <- s_fe^2 - s_re^2
102     if (var_diff > 0) {
103       t_stat <- (b_fe - b_re) / sqrt(var_diff)
104       cat(sprintf("Variable: %-12s | t-stat: %8.4f | Significant (5%): %s\n",
105                 var, t_stat, abs(t_stat) > 1.96))
106     } else {
107       cat(sprintf("Variable: %-12s | Variance diff <= 0 (Cannot compute t-stat)\n", var))
108     }
109   }
110 }

```

Calculated t-statistics comparing FE and RE coefficients:

- small: 1.146
- aide: 0.128
- tchexper: -1.938 (close to significance)
- white_asian: 1.218
- freelunch: -0.096

None of these are statistically significant at the 5% level ($|t| < 1.96$), implying that individually, we do not reject the null of exogeneity for these variables.

When applied to the "boy" coefficient, the variance difference (Var_FE - Var_RE) is negative, meaning the t-statistic cannot be computed. This is a known finite-sample issue with the Hausman test and usually suggests the estimators are so close that sampling error dominates, or that the RE model is misspecified.

f. Mundlak Approach.

```

12 # f. Mundlak Approach
13 cat("\n15.20f: Mundlak Approach\n")
14 # Create school-level averages
15 star_mundlak <- star %>%
16   group_by(schid) %>%
17   mutate(
18     mean_small = mean(small, na.rm = TRUE),
19     mean_aide = mean(aide, na.rm = TRUE),
20     mean_tchexper = mean(tchexper, na.rm = TRUE),
21     mean_boy = mean(boy, na.rm = TRUE),
22     mean_white_asian = mean(white_asian, na.rm = TRUE),
23     mean_freelunch = mean(freelunch, na.rm = TRUE)
24   ) %>%
25   ungroup()
26
27 # Fit OLS model including the means
28 mod_mundlak <- lm(readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
29   mean_small + mean_aide + mean_tchexper + mean_boy + mean_white_asian + mean_freelunch,
30   data = star_mundlak)
31
32 cat("Mundlak Regression Results (OLS):\n")
33 print(summary(mod_mundlak))
34
35 # Test joint significance of the mean variables
36 mean_vars <- c("mean_small", "mean_aide", "mean_tchexper", "mean_boy", "mean_white_asian", "mean_freelunch")
37
38 mundlak_test <- linearHypothesis(mod_mundlak, mean_vars, test = "F")
39 cat("\nJoint F-Test for School Averages (Mundlak Test):\n")
40 print(mundlak_test)
41

```

We created school-averages for all explanatory variables and added them to an OLS model. We then performed a joint F-test on these school-averages (mean_small, mean_aide, mean_tchexper, mean_boy, mean_white_asian, mean_freelunch).

The joint F-test statistic is 21.004 ($p < 0.0001$). Because the school averages are jointly significant, we reject the null hypothesis of no endogeneity. This confirms that the unobserved school heterogeneity IS correlated with the explanatory variables, meaning the Fixed Effects model is the appropriate choice over Random Effects.